Junyi (Conny) Zhou

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EDUCATION

- **Harvard University** — Boston, MA
  M.S. in Health Data Science — *August 2025 – December 2026*
  - Coursework: Deep Learning, Healthcare Machine Learning
- **Emory University, College of Arts and Sciences** — Atlanta, GA
  B.S. in Applied Mathematics and Statistics; Minor in Computer Informatics; GPA: 3.925 — *August 2021 – May 2025*
  - Honors: Dean's List Spring 2022, Spring 2023
  - Coursework: Probability and Statistics, Machine Learning, NLP, Data Structures, Database Systems

PROGRAMMING SKILLS

- **Languages**: Python, C++, Java, SQL, R, JavaScript
- **ML Frameworks**: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, Hugging Face Transformers
- **Databases**: MySQL, PostgreSQL, MongoDB, Pinecone (Vector DB), Hadoop, Spark
- **Tools**: Claude Code, Cursor, Git, Docker, Kubernetes, Jupyter
- **Cloud**: AWS (SageMaker, Lambda, EC2, S3, Redshift), Snowflake, BigQuery
- **Certifications**: AWS Certified Cloud Practitioner, AWS Certified AI Practitioner, AWS Machine Learning Associate

RESEARCH PROJECTS

- **Airway Management Simulation Chatbot** — Atlanta, GA
  Full-Stack Engineer, Scrum Master — Emory Pediatric Hospital — *January 2024 – Present*
  - Architected a clinical **agentic LLM pipeline** via a **RAG** system and **OpenAI models**, minimizing hallucinations and ensuring high data privacy for simulations.
  - Built a **HIPAA-compliant** full-stack platform (**Python/Flask**, React, MongoDB, AWS) tracking resident performance metrics via secure, real-time analytics.
  - Implemented **Prompt Engineering** and context window management to drive dynamic scenario generation for rapid cycle deliberate practice clinical training.
  - Led an Agile team of 6, facilitating code reviews and integrating clinician feedback to continually refine the NLP pipeline. [1]
- **Cross-Species Drug Translation using Optimal Transport & Generative AI** — Boston, MA
  Graduate Researcher — Mooney Lab & Alvarez-Melis Lab, Wyss Institute — *February 2026 – Present*
  - Spearheaded computational frameworks using **Optimal Transport**, **VAEs**, and **diffusion models** to predict animal-to-human clinical trial translation via scRNA-seq atlases.
  - Engineered a custom **PyTorch** CellOT implementation from scratch, modeling complex species and treatment effects directly within VAE latent spaces.
  - Designed **scanpy** pipelines for automated cell-type labeling, integrating massive public and lab-generated scRNA-seq datasets for cancer and autoimmunity applications.
  - Trained out-of-distribution models mapping cellular subpopulations (e.g., regulatory vs. helper T cells) across species, evaluating GENOT to optimize T cell matching.
- **AlphaFold Protein–Nucleic Acid Interaction Prediction** — Atlanta, GA
  Data Analyst — Bou-Nader Lab, Emory Dept. of Biochemistry — *February 2025 – August 2025*
  - Engineered a pipeline integrating **AlphaFold-Multimer** and **AlphaFold3** architectures to predict complex protein–nucleic acid interactions using **MSA** data.
  - Automated high-throughput inference on **HPC** clusters with **parallel scheduling** to efficiently process structural predictions across GPU nodes.
  - Developed **Python** scripts (PyMOL, ChimeraX) filtering predicted structures via **pLDDT** and **PAE** scores, selecting top-performing complexes for wet-lab validation.
  - Authored CLI tools and technical documentation to democratize advanced ML inference tools for non-technical researchers. [2]
- **Reimplementation of Transformer Architecture** — Atlanta, GA
  Machine Learning Researcher — Emory Dept. of Computer Science — *October 2024 – January 2025*

[1] https://d3g1qw60hz7yw7.cloudfront.net
[2] https://github.com/JunyiZhou-Conny/AlphaFold-Bou-Nader-Lab

- Implemented a Transformer in **PyTorch** from scratch, building **Multi-Head Self-Attention**, **Positional Encodings**, and **Layer Normalization** to reproduce benchmarks.
- Conducted extensive hyperparameter tuning and optimized the **Cross-Entropy Loss** function, achieving a 9.38 BLEU score on English-to-German translation.
- Benchmarked custom implementation against standard `nn.Transformer` modules to strictly validate architectural correctness and training efficiency. [3]

[3]https://github.com/JunyiZhou-Conny/Reimplementation-of-Transformer